RAFFLES INSTITUTION
H2 Mathematics (9758)
2026 Year 6

Additional Practice Questions for Chapter 9C: Differential Equations

1 Solve the following differential equations.

(a) $\dfrac{1}{y^2}+\dfrac{\mathrm{d}y}{\mathrm{d}x}=1,\ y>1,$

(b) $\dfrac{\mathrm{d}y}{\mathrm{d}x}=\mathrm{e}^{x+y}$, given that $y=0$ when $x=0$.

2 Use the substitution $y=ux^2$ to show that the differential equation

$$\left(x^2-2y\right)\frac{\mathrm{d}y}{\mathrm{d}x}-2\left(2x^2+y\right)x=0 \text{ for } x\neq 0,$$

can be reduced to $x\dfrac{\mathrm{d}u}{\mathrm{d}x}=\dfrac{4+4u^2}{1-2u}$.

Hence show that $\ln\left[A\left(x^4+y^2\right)\right]=\tan^{-1}\left(\dfrac{y}{x^2}\right)$, where A is a positive constant.

3 For any one kind of glass, the intensity I of light transmitted through glass varies with the thickness x of the glass. It is given the decrease of I with respect to x is directly proportional to x. If $I=10$ when $x=0$, find I as a function of x. If I is reduced from 10 to 7 after passing through glass 5 cm thick, find the intensity of light from the same source after passing through glass 6 cm thick. [7]

4 At each instant of time, the rate of increase of money in a savings account is proportional to the amount in the account at that instant. The constant of proportionality does not vary with time. Denote the amount in the account at time t years by $\$x$. When $x=1000$, the rate of increase is \$50 per year. Obtain a differential equation relating x and t.

Initially, when $t=0$, the account contained \$900.

(i) Find the amount in the account exactly 3 years later.

(ii) Find, in years correct to 2 decimal places, the time when the account contains \$1800.

5 Booboo bird is an endangered species of birds on Cook's Island. The ornithologist found that the population of the Booboo birds is decreasing at a rate inversely proportional to the number of birds and decides to bring in a fixed number of the Booboo birds yearly.

At time t years after Year 2000, x is the number of Booboo birds on the island. It is also known that the number of birds remain constant when $x = 20$.

Show that $\frac{\mathrm{d}x}{\mathrm{d}t} = k\left(1 - \frac{20}{x}\right)$ where k is a positive constant. [2]

The ornithologist found 104 and 121 Booboo birds on Cook's Island at the end of Years 2005 and 2006 respectively. Use the above model to predict the number of Booboo birds on the island at the end of Year 2015. State an assumption made in your calculation. [5]

6 A race called the Matrices live on an isolated island called Vector. The number of births per unit time is proportional to the population at any time t and the number of deaths per unit time is proportional to the square of the population.

If the population at time t is P, show that $\frac{\mathrm{d}P}{\mathrm{d}t} = aP - bP^2$, where a and b are positive constants.

Solve the differential equation for P in terms of t, given that $P = \frac{2a}{3b}$ when $t = 0$.

Show that there is a limit to the size of the population.

7 A tank initially contains 400 litres of solution with 100 kg of salt dissolved in it. A solution containing 0.125 kg of salt per litre flows into the tank at a rate of 12 litres per minute and the solution flows out at the same rate. You should assume that the inflow is instantaneously and thoroughly mixed with the contents of the tank. If the amount of salt in the tank is q kg at the end of t minutes, show that

$$\frac{\mathrm{d}q}{\mathrm{d}t} = 1.5 - 0.03q\,.$$ [2]

Find the time taken for the concentration of salt in the tank to reach 0.16 kg per litre. [5]
(Concentration of salt = the amount of salt per unit volume of solution in the tank.)

State what happens to q for large values of t. Sketch a graph of q against t. [3]

8 In a chemical reaction in which a compound X is formed from a compound Y and other agents the masses of X and Y present at time t are x and y respectively. The sum of the two masses is constant and at any time the rate at which x is increasing is proportional to the product of the two masses at that time. Show that the equation governing the reaction is of the form $\frac{\mathrm{d}x}{\mathrm{d}t} = kx(a - x)$ and interpret the constant a.

If $x = 0.1a$ at time $t = 0$ find, in terms of k and a, the time at which the mass of Y is $0.1a$. [9]

9 At the intensive care unit of a hospital, patients of a particular condition receive a certain treatment drug through an intravenous drip at a constant rate of 30mg per hour. Due to the limited capacity for absorption by the body, the drug is lost from a patient's body at a rate proportional to x , where x is the amount of drug (in mg) present in the body at time t (in hours). It is assumed that there is no presence of the drug in any patient prior to admission to the hospital.

(i) Form a differential equation involving x and t and show that $x = \frac{30}{k}\left(1 - \mathrm{e}^{-kt}\right)$ where k is a positive constant. [4]

(ii) If there is more than 1000mg of drug present in a patient's body, it is considered an overdose. Suppose the drug continues to be administered, determine the range of values of k such that a patient will have an overdose. [2]

(iii) For a particular patient, $k = \frac{1}{50}$. Find the time required for the amount of the drug present in the patient's body to be 200mg. [3]

10 A toy paratrooper is dropped from a building and the attached parachute opens the moment it is released. The toy drops vertically and the distance it drops after t seconds is x metres. The motion of the toy can be modelled by the differential equation

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + k\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 = 10,$$

where k is a constant.

By substituting velocity, $v = \frac{\mathrm{d}x}{\mathrm{d}t}$, write down a differential equation in v and t. [1]

Given that $\frac{\mathrm{d}v}{\mathrm{d}t} = 6$ when $v = \sqrt{10}$, and that the initial velocity of the toy is zero, show that

$$v = \frac{5(1 - \mathrm{e}^{-4t})}{1 + \mathrm{e}^{-4t}},$$

and deduce the velocity of the toy in the long run. [6]

The toy is released from a height of 10 metres. Find the time it takes for the toy to reach the ground. [5]

11 At time $t = 0$, a boiled potato is taken from a pot on a stove and left to cool in a kitchen. The internal temperature of the potato is 91 degrees Celsius ($°C$) at time $t = 0$, and the internal temperature of the potato is greater than $27\,°C$ for all times $t > 0$. The internal temperature of the potato at time t minutes can be modelled by the function H that satisfies the differential equation $\frac{\mathrm{d}H}{\mathrm{d}t} = -\frac{1}{4}(H - 27)$ where $H(t)$ is measured in degrees Celsius and $H(0) = 91$.

(i) Write an equation for the tangent to the graph of H at $t = 0$. Use this equation to approximate the internal temperature of the potato at time $t = 3$.

(ii) Use $\frac{\mathrm{d}^2H}{\mathrm{d}t^2}$ to determine whether your answer in **(i)** is an underestimate or overestimate of the internal temperature of the potato at time $t = 3$.

(iii For $t < 10$, an alternative model for the internal temperature of the potato at time t minutes is the function G that satisfies the differential equation $\frac{\mathrm{d}G}{\mathrm{d}t} = -(G - 27)^{\frac{2}{3}}$ where $G(t)$ is measured in degrees Celsius and $G(0) = 91$. Find an expression for $G(t)$.
Based on this model find the internal temperature of the potato at time $t = 3$.

12 A conservationist is studying a small population of a species of rodent. The number of rodents, N, is modelled by the differential equation

$$\frac{\mathrm{d}N}{\mathrm{d}t} = aN(N - b),$$

where a and b are suitably chosen constants and t is the time in weeks from the start of the investigation.

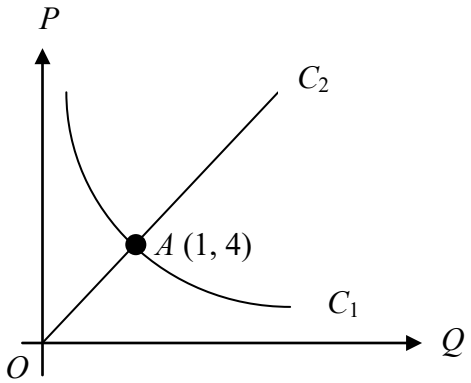
Constants are chosen for the model so that it gives 40 for the initial number of rodents and 400 for the number in the long-term.
Find an expression for N in terms of a and t. Explain why $a < 0$.

If in addition, the model predicts that the rate of growth of the population will be maximum after $10 \ln 3$ weeks, find a.

13 In the diagram below, the curve C_1 and the line C_2 illustrate the relationship between price (P dollars per kg) and quantity (Q tonnes) for consumers and producers respectively.

The curve C_1 shows the quantity of rice that consumers will buy at each price level while the line C_2 shows the quantity of rice that producers will produce at each price level. C_1 and C_2 intersect at point A, which has the coordinates (1, 4).

The quantity of rice that consumers will buy is inversely proportional to the price of the rice. The quantity of rice that producers will produce is directly proportional to the price.



(i) Interpret the coordinates of A in the context of the question. [1]

(ii) Solve for the equations of C_1 and C_2, expressing Q in terms of P. [2]

Suggest a reason why producers might use $P = aQ + b$, where $a, b \in \mathbb{R}^+$, instead of C_2 to model the relationship between price and quantity of rice produced. [1]

Shortage occurs when the quantity of rice consumers will buy exceeds the quantity of rice producers will produce. It is known that the rate of increase of P after time t months is directly proportional to the quantity of rice in shortage.

(iii) Given that the initial price is \$3 and that after 1 month, the price is \$3.65, find P in terms of t and sketch this solution curve, showing the long-term behaviour of P. [7]

THE END